

Weekly Progress Log – Week 1

Project Title: Data Science with Python

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Week: 1

Objectives for Week

- Set up a complete Python development environment.
- Install and configure Python, Jupyter Notebook, and VS Code.
- Create and activate a Python virtual environment.
- Become familiar with core Python concepts including data types, lists, dictionaries, loops, functions, file handling, and basic object-oriented programming.
- Complete beginner-level Python programming exercises.
- Develop three small Python programs: a calculator, a simple to-do list, and a basic file parser.
- Learn the basics of Git and GitHub.
- Create and organize a GitHub repository for weekly internship work.
- Prepare documentation for environment setup and project dependencies.

Work Completed

- Installed and configured Python, Jupyter Notebook, and Visual Studio Code.
- Created a virtual environment named 'venw' for the project.
- Installed the required Python libraries including:
 - ✓ NumPy
 - ✓ Pandas
 - ✓ Matplotlib
 - ✓ Scikit-learn
 - ✓ Jupyter
 - ✓ IPyKernel
- Created a 'requirements.txt' file to document project dependencies.

- Created a GitHub repository named 'DataScience_with_Python'.
- Prepared the main 'README.md' containing setup instructions and repository structure.
- Completed beginner Python exercises covering:
 - ✓ Variables and data types
 - ✓ Lists
 - ✓ Dictionaries
 - ✓ Conditional statements
 - ✓ Loops
 - ✓ Functions
 - ✓ File handling
 - ✓ Basic object-oriented programming
- Completed additional Python practice exercises using Jupyter Notebook.
- Developed three small Python programs:
 - ✓ Calculator
 - ✓ Simple To-Do List
 - ✓ Basic File Parser
- Used a text file ('students.txt') for file-handling and file-parser practice.
- Organized Week 1 files inside the GitHub repository.

Challenges Faced

- Understanding how Python virtual environments work and why the environment folder should not be uploaded directly to GitHub.
- Understanding the purpose of the 'requirements.txt' file and deciding which packages should be included.
- Learning how GitHub repository structure should be organized for weekly internship work.
- Understanding the difference between local project files and files that should be uploaded to GitHub.
- Becoming familiar with cloning a GitHub repository and environment setup instructions.
- Managing file paths and virtual environment activation commands on Windows.
- Applying file-handling concepts while creating the basic file parser.

Solutions Implemented

- Created a separate Python virtual environment to keep project dependencies isolated.

- Documented the steps required to recreate the environment in the main 'README.md'.
- Created a 'requirements.txt' file containing the main libraries required for the project.
- Organized the repository into separate weekly folders to keep the project structured.
- Used Jupyter Notebook to practice and test Python concepts interactively.
- Used clear filenames to separate exercises, practice work, scripts, and supporting files.
- Reviewed GitHub workflow and repository organization before uploading the Week 1 work.
- Practiced file reading and processing using 'students.txt'.

Individual Tasks

Responsibility	Description of Contribution	Remark
Environment Setup	Installed Python, Jupyter Notebook, VS Code, and configured the Python virtual environment 'venw'.	Development environment successfully configured.
Dependency Management	Verified installed Python libraries and created 'requirements.txt'.	Dependencies documented for environment recreation.
Python Fundamentals	Completed beginner exercises covering data types, lists, dictionaries, loops, functions, file handling, and basic OOP.	Improved understanding of core Python concepts.
Python Practice	Completed additional practice exercises using Jupyter Notebook.	Strengthened Python programming skills.
Calculator Program	Developed a basic calculator for arithmetic operations.	Successfully implemented basic functions and user input.
To-Do List Program	Developed a simple to-do list application.	Practiced lists, loops, conditions, and functions.
File Parser	Developed a basic file parser using students.txt.	Practiced reading and processing text files.
GitHub Repository	Created and organized the 'DataScience_with_Python' GitHub repository.	Week 1 work organized and prepared for submission.
Documentation	Created the main README with setup instructions and repository structure.	Project setup can be understood and reproduced by others.